



Enhanced Financial Portfolio Optimization with Risk Management using the GraphSAGE-PPO model

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Abstract

Financial portfolio management is a longstanding yet increasingly complex problem in modern finance, characterized by high volatility, dynamic inter-asset dependencies, and the need for timely decision-making. Traditional methods often rely on static models and expert-driven heuristics, which struggle to adapt to changing market conditions and require significant human oversight. Moreover, many existing machine learning (ML) and deep learning (DL) approaches inadequately capture the relational structure among financial assets, leading to suboptimal portfolio allocation strategies. This study addresses these challenges by proposing a fully autonomous, data-driven portfolio optimization framework that eliminates the need for manual intervention. The study introduces a novel hybrid model that integrates GraphSAGE, a graph neural network designed to learn rich, scalable node representations, with the Proximal Policy Optimization (PPO) algorithm, a state-of-the-art reinforcement learning (RL) method. By modeling financial markets as graphs—where nodes represent individual assets and edges encode relationships such as correlation or co-movement—our approach effectively captures both the individual features of assets and their structural interactions. This model outperforms traditional and baseline ML strategies by delivering higher risk-adjusted returns, better market shock resilience, and superior diversification. This graph-RL fusion offers a powerful, explainable, and scalable solution for modern portfolio management.



1 Introduction

In both individual and institutional finance, constructing an effective financial portfolio is essential for achieving long-term investment objectives. A well-diversified portfolio distributes risk across various asset classes—such as equities, bonds, and commodities—thereby minimizing the likelihood that a single adverse market event will result in significant financial loss (Frahm & Wiechers, 2011). It also enables investors to align their strategies with personal risk tolerance and investment horizons. In today's increasingly volatile and data-driven financial landscape, the significance of thoughtful portfolio structuring has intensified. The ability to design portfolios that are both resilient and return-optimized is now considered a cornerstone of sound financial planning (Sanjiv Das, 2010). As markets grow more interconnected and dynamic, intelligent portfolio construction becomes imperative for maintaining financial stability and capturing emerging opportunities. Portfolio optimization refers to the process of allocating capital across assets in a manner that seeks to maximize expected returns while managing

associated risks. The classical foundation for this task was laid by Markowitz's Modern Portfolio Theory, which introduced the trade-off between expected return and portfolio variance [Harry \(1952\)](#). However, real-world financial environments are considerably more complex. Risk arises from multiple sources—including market volatility, liquidity constraints, and regime changes—and often has nonlinear effects on performance. Consequently, risk management has become an integral component of modern portfolio optimization ([Fabozzi et al., 2007](#); [DeMiguel et al., 2009](#)). To better reflect these realities, contemporary models often incorporate additional risk constraints, such as Conditional Value-at-Risk (CVaR) ([Rockafellar & Uryasev, 2000](#); [Sven S. Groth a, 2011](#)).

Despite the value of traditional frameworks such as the Capital Asset Pricing Model (CAPM) [Sharpe \(1964\)](#) and the Black-Litterman model [Black & Litterman \(1992\)](#), these approaches are built on assumptions—such as normally distributed returns and fixed asset correlations—that often fail in practice ([Avramov et al., 2006](#)). In recent years, advances in artificial intelligence, particularly deep reinforcement learning, have opened new avenues for creating adaptive, data-driven portfolio strategies ([Culkin & Das, 2017](#); [Thomas Fischer, 2018](#)). Yet, many DRL algorithms such as Proximal Policy Optimization, Deep Deterministic Policy Gradient, and Advantage Actor-Critic typically model each asset independently, ignoring the evolving interrelationships among them ([Yang, 2023](#)). This limitation restricts their ability to perform reliably in volatile or structurally shifting markets. Moreover, conventional machine learning and deep learning techniques struggle with the high noise, non-stationarity, and lack of relational context inherent in financial datasets ([Jang & Seong, 2023](#)).

Graph Neural Networks, and in particular the GraphSAGE architecture, offer promising capabilities for embedding the relational structure of asset networks into financial modeling ([Cheng et al., 2022](#); [Fuli Feng, 2019](#)). However, the integration of GNNs into DRL-based portfolio frameworks remains limited. Additionally, many existing models focus primarily on return or Sharpe ratio maximization and fail to incorporate mechanisms for managing downside risk during periods of market stress ([Xinyao Wang, 2025](#)). This research proposes a novel hybrid framework that leverages GraphSAGE to model asset relationships, employs PPO for adaptive allocation, and incorporates a risk-aware reward function to improve robustness and resilience.

The primary objective of this study is to develop a portfolio optimization framework that combines GraphSAGE's structural representation capabilities with PPO's reinforcement learning-based decision-making under uncertainty. The model is trained on historical financial data, with feature relevance enhanced using SHAP (SHapley Additive exPlanations) to improve transparency and model interpretability ([Scott M. Lundberg, 2017](#); [Maity et al., 2022](#)). A custom reward function is employed to balance profitability with volatility, thereby promoting more stable investment behavior ([Moody & Saffell, 2001](#)). The proposed model is benchmarked against traditional strategies such as Equal Weight and the S&P 500 index ([Fama & French, 1993](#); [Garlappi et al., 2007](#)). A distinguishing feature of the framework is the inclusion of a fallback mechanism that activates an Equal Weight strategy when the learned policy underperforms during volatile conditions, thereby preserving capital. The significance of this research lies in its potential to address several critical gaps in modern portfolio optimization. Traditional financial models often rely on oversimplified assumptions that fail under real-world volatility, while many machine learning approaches lack interpretability and ignore asset interdependencies. At the same time, the increasing complexity and speed of financial markets demand models that are not only adaptive but also robust to sudden regime shifts and downturns. By integrating GraphSAGE for relational awareness, PPO for adaptive allocation, SHAP for explainability, and fallback mechanisms for risk containment, this study presents a holistic framework suited to contemporary financial challenges. The proposed model enhances decision transparency, supports real-time responsiveness, and prioritizes both return and resilience—qualities that are essential for institutional adoption and long-term financial planning in uncertain markets.

The remainder of this paper is organized as follows. Section 2 provides a detailed review of existing literature in classical portfolio theory, machine learning for finance, and graph-based portfolio models. Section 3 describes the architecture of the proposed system, including the GraphSAGE encoder and PPO policy agent. Section 4 outlines the data sources, preprocessing steps, and baseline strategies used for empirical evaluation. Section 5 presents experimental results and performance comparisons. Finally, Section 6 concludes the paper and discusses directions for future research.

2 Literature Review

2.1 Reinforcement Learning in Portfolio Optimization

Reinforcement learning has emerged as a transformative approach in financial portfolio optimization, primarily due to its ability to autonomously learn adaptive investment strategies in highly dynamic and uncertain environments. Unlike traditional rule-based or static optimization methods, RL agents interact with the market environment, receiving feedback in the form of rewards and iteratively improving their decision policies. Early foundational work by [Moody & Saffell \(2001\)](#) demonstrated the effectiveness of policy-gradient methods in trading systems, specifically by directly maximizing the Sharpe ratio rather than relying on fixed, predefined trading rules. This approach opened the door to more flexible and data-driven portfolio management strategies. Building on this, [Zhipeng Liang \(2018\)](#) explored adversarial training methods and conducted comparative analyses of various RL algorithms, such as Deep Deterministic Policy Gradient, Proximal Policy Optimization, and Policy Gradient, showing that these models often outperformed conventional approaches, particularly during periods of high market volatility. Further, [Srijan Sood \(2023\)](#) benchmarked deep RL agents against the classic mean-variance optimization framework and found that RL-based models consistently achieved superior risk-adjusted returns across different market regimes, highlighting their adaptability. However, despite their promise, most RL-based approaches in finance tend to overlook the intricate structural relationships between assets and typically employ reward functions that focus solely on profit maximization. This narrow focus can render such models unstable and less robust in real-world markets, where asset correlations and risk profiles are subject to frequent and unpredictable fluctuations.

2.2 Graph Neural Networks in Financial Modeling

Graph Neural Networks have recently garnered significant attention in financial modeling due to their capacity to represent and exploit the complex relational structures that exist between financial assets. In contrast to traditional machine learning models that treat assets as independent entities, GNNs encode the dependencies and interactions among assets as edges in a graph, allowing for richer and more context-aware feature representations. [Jianian Wang \(2022\)](#) provided reviews that GNNs could effectively capture hidden dependencies between stocks, leading to more stable and robust deep learning models for portfolio construction. Similarly, [Kamesh Korangi \(2024\)](#) introduced a dynamic graph attention framework that leveraged thirty years of mid-cap stock data, illustrating that relational models could better capture market behaviors and outperform non-relational baselines. The integration of graph convolutional layers with reinforcement learning, as seen in systems like DeepPocket ([Soleymani F., 2021](#)), further enables models to dynamically update their understanding of the market as new data arrives. Despite these advancements, most GNN-based models in finance are designed as static feature extractors and are rarely integrated with real-time, interactive decision-making processes. This limits their practical utility, as there remains a gap between structural market understanding and the ability to make timely, actionable investment decisions.

2.3 Hybrid Graph and Reinforcement Learning Approaches

The intersection of graph-based modeling and reinforcement learning has given rise to a new class of hybrid approaches that seek to combine the strengths of both paradigms. These models aim to leverage the relational insights provided by GNNs while simultaneously harnessing the adaptive decision-making capabilities of RL agents. For example, [Sun et al. \(2024\)](#) proposed a hybrid GraphSAGE-PPO model in which the graph structure among assets was learned and updated as an integral part of the RL training pipeline, thereby providing the RL agent with a more context-aware and dynamic view of asset relationships for improved asset allocation. Other notable efforts, such as DeepPocket and dynamic graph attention models ([Soleymani F., 2021](#); [Kamesh Korangi, 2024](#)), have explored similar integrations, but often with limitations. In many cases, the graph structure is either fixed or updated manually, which restricts the model's adaptability to fast-changing market conditions. Moreover, the reward functions in these hybrid systems are frequently designed to maximize returns without explicitly penalizing

undesirable outcomes such as high volatility or large drawdowns. As a result, while the combination of GNNs and RL holds significant promise for portfolio optimization, current implementations often lack the flexibility, robustness, and risk sensitivity required for deployment in high-frequency or high-risk trading environments.

2.4 Machine Learning and Deep Learning Approaches in Portfolio Management

Beyond RL and GNNs, traditional machine learning and deep learning techniques have also played a substantial role in advancing portfolio modeling and prediction tasks. Classical ML models such as random forests, support vector machines, and gradient boosting machines have been widely used to forecast asset returns, volatility, and market trends, often outperforming linear models in their ability to handle noisy and nonlinear financial data (J. B. Heaton, 2017; Zhang, 2019). In the realm of deep learning, Thomas Fischer (2018) applied long short-term memory networks to predict stock price movements on the S&P 500, demonstrating a consistent edge over benchmark strategies. Similarly, Tsantekidis Avraam (2017) utilized a combination of convolutional neural networks and LSTMs on limit order book data to model both spatial and temporal patterns, resulting in improved short-term trading accuracy. Despite their strong predictive capabilities, these ML and DL approaches often stop at the prediction stage and are not directly integrated into portfolio optimization workflows. Additionally, their black-box nature and limited adaptability to rapidly changing market conditions reduce their practical usability for building resilient and robust portfolios, especially during periods of financial stress.

2.5 Model Explainability and SHAP Interpretability in Finance

As portfolio optimization models grow in complexity—especially with the adoption of deep reinforcement learning and graph-based neural architectures—their interpretability becomes increasingly limited. Most RL or GNN-based models behave as "black-boxes," making it difficult to understand the rationale behind asset allocation decisions (Widiputra et al., 2021). For example, the deep ensemble trading system proposed by Zhang et al. Rundong Wang (2020) integrates convolutional and recurrent layers for multi-asset forecasting but offers minimal insight into which features drive predictions. To address this challenge, model-agnostic explainability frameworks like SHapley Additive exPlanations have emerged. SHAP provides localized feature attributions based on game-theoretic principles, making it particularly valuable in finance where transparency is critical. By integrating SHAP into our GraphSAGE-PPO model, we ensure that each portfolio decision can be decomposed into feature-level contributions, improving accountability and aiding compliance in regulated environments. Moreover, this integration helps validate learned representations and detect data leakage or overfitting by revealing spurious correlations in model behavior.

2.6 Research Gaps and Contributions

A synthesis of the literature reveals several persistent gaps that motivate the present study. RL models, while effective in learning adaptive trading strategies, frequently ignore the underlying structure and relationships between assets, leading to suboptimal risk management. GNN-based models, although adept at capturing asset relationships, are typically not designed for active decision-making in dynamic market settings. Hybrid approaches that combine GNNs and RL have shown potential, but they often rely on static graphs and employ reward functions that are insensitive to risk, focusing primarily on return maximization. Even the most advanced ML and DL models, despite their predictive prowess, fall short when it comes to integration with portfolio-level objectives and real-time optimization. To address these limitations, this research introduces a unified framework that combines dynamic graph learning with a reinforcement learning agent, enhanced by a custom reward function that penalizes drawdowns and volatility while promoting portfolio stability. Furthermore, the model incorporates a fallback mechanism that reverts to an equal-weight allocation during periods of extreme market conditions, thereby ensuring greater robustness and capital preservation. Through this integrated approach, the proposed work aims

to bridge the gaps between structural modeling, predictive learning, and real-time portfolio optimization, offering a comprehensive solution for modern financial markets. Summary of Related Works and Their Relevance to the Proposed GraphSAGE-PPO Framework given in Table 1.

Table 1: Summary of Related Works and Their Relevance to the Proposed GraphSAGE-PPO Framework

Paper Reference	Objective	Method Used	Dataset	Limitations	Contribution of Present Work
(Moody & Saffell, 2001)	Maximize Sharpe ratio using RL for trading	Policy Gradient, Sharpe Ratio-based reward	Historical trading data	Ignores asset relationships; limited to reward-based optimization	Adds dynamic asset structure modeling via GNN and risk-aware rewards
(Zhipeng Liang, 2018)	Compare RL algorithms under adversarial settings	PPO, DDPG, PG with adversarial training	Simulated and market data	Focused on algorithmic robustness; lacks inter-asset structure	Integrates relational structure with RL via dynamic GNN
(Srijan Sood, 2023)	Benchmark RL models against mean-variance portfolio theory	Deep RL agents	Real-world financial market data	Focus on return maximization; weak risk control	Introduces volatility and drawdown penalties into reward function
(Kamesh Korangi, 2024)	Use dynamic graph attention on 30-year mid-cap stock data	Graph Attention Networks	Mid-cap stocks over 30 years	No agent integration; graph guides representation only	Joint GNN-RL learning for active portfolio allocation
(Soleymani F., 2021)	Propose hybrid GNN-RL model (DeepPocket)	GNN + RL Hybrid	Financial time series data	Static graphs; lacks risk-sensitive reward design	Introduces dynamic graph learning and custom reward shaping
(Sun et al., 2024)	Combine GraphSAGE with PPO for allocation decisions	GraphSAGE + PPO	Historical financial data	Graph often fixed; reward lacks risk-awareness	Adds risk-penalizing reward and equal-weight fallback strategy
(J. B. Heaton, 2017)	Compare ML models with financial baselines	Random Forests, SVM, Gradient Boosting	Asset return and volatility data	Not used in active portfolio optimization	Integrates ML output into real-time RL-based allocation
(Thomas Fischer, 2018)	Predict S&P 500 stock prices using LSTM	LSTM deep learning model	S&P 500 historical data	Purely predictive; not used for portfolio decision-making	Closes prediction-to-decision loop using RL integration
(Widiputra et al., 2021)	Forecast multiple assets using ensemble deep models	CNN + LSTM ensemble	Multi-asset data	Black-box model; lacks explainability	Uses SHAP for transparent model attribution and debugging

3 Methodology

This study extends previous research that introduced a hybrid GraphSAGE-PPO framework for financial portfolio optimization by incorporating explainable AI mechanisms and fallback controls (Sun et al., 2024). The methodology follows a systematic and modular approach, integrating diverse financial data sources and state-of-the-art machine learning techniques to tackle the multifaceted challenges inherent in portfolio management.

The process begins with the collection of a broad spectrum of financial time series data, including asset prices, technical indicators, macroeconomic variables, and global market indices. These datasets are obtained from publicly available APIs such as `yfinance`, `stockstats`, and the Federal Reserve Economic Data (FRED). Once acquired, the data undergoes comprehensive preprocessing, which includes normalization and forward-filling techniques to handle missing values. To enhance both computational efficiency and interpretability, SHAP values are employed for feature selection. This step enables the model to prioritize the most informative indicators across asset-level and macroeconomic dimensions.

To represent the structural relationships within financial markets, a graph-based model is constructed in which nodes denote individual assets or indices and edges represent statistically significant correlations. This graph, enriched with the selected feature matrix, serves as the input to a Graph Neural Network. Specifically, the GraphSAGE architecture is utilized to learn low-dimensional embeddings for each node, effectively capturing intrinsic asset properties as well as inter-asset dependencies (Sun et al., 2024). These embeddings are then passed to a Proximal Policy Optimization agent, which is trained in a simulated trading environment. The PPO agent continuously adjusts portfolio weights in response to changing market dynamics, with the objective of maximizing risk-adjusted returns. To enhance robustness, a fallback mechanism is incorporated into the system. If the PPO agent experiences a predefined sequence of consecutive losses, the model automatically switches to an equal-weight allocation strategy. This conservative failsafe is designed to mitigate potential compounding losses during periods of heightened volatility or market instability. The model's performance is evaluated using well-established financial metrics, including cumulative return, Sharpe ratio, skewness, and win ratio. These evaluation criteria facilitate rigorous comparisons with traditional portfolio management strategies. By combining graph-based representation learning, reinforcement learning, explainability through SHAP, and embedded risk mitigation strategies, the proposed methodology offers a comprehensive, adaptive, and resilient framework for modern portfolio optimization.

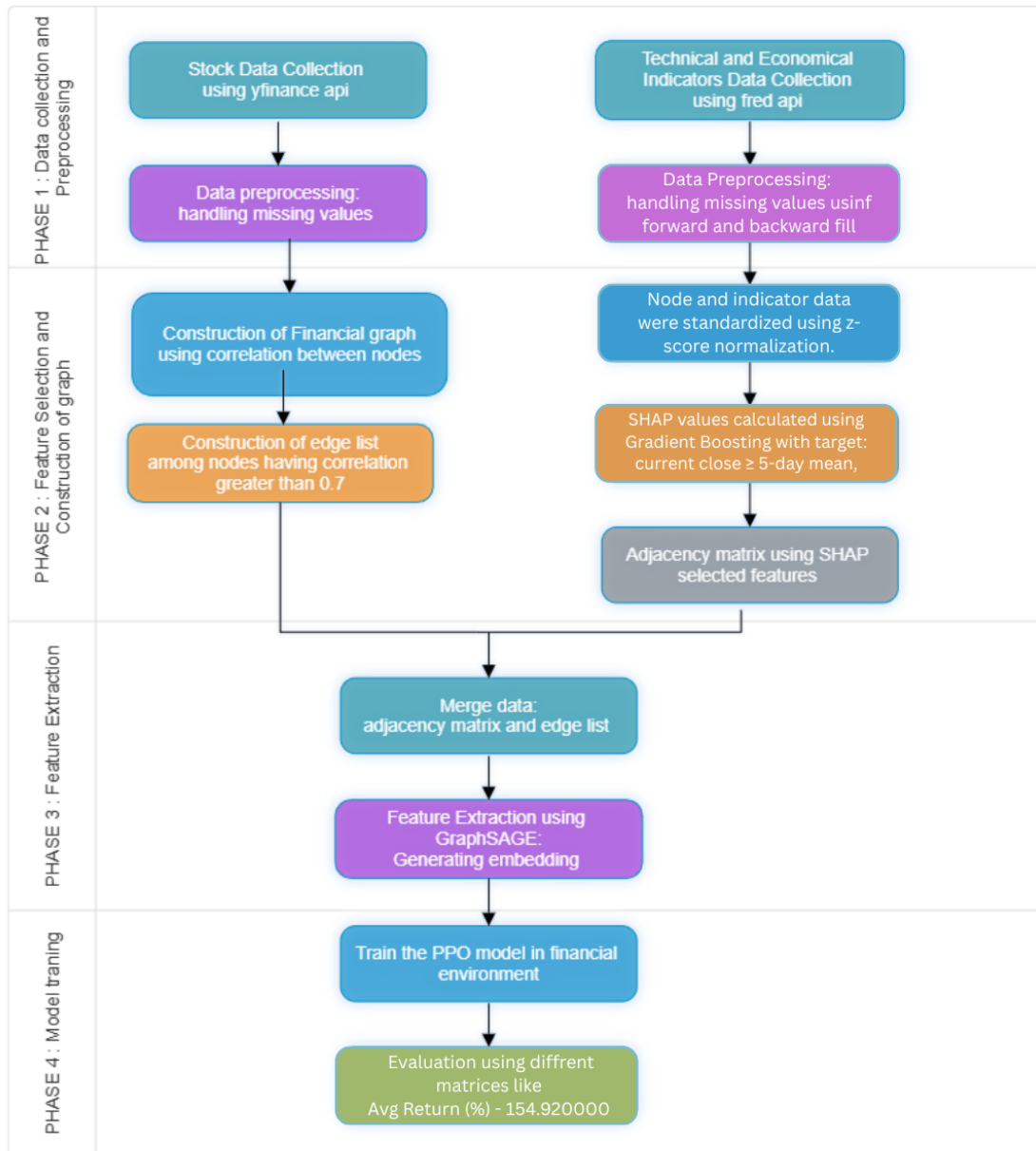


Figure 1: Methodology Workflow for GraphSAGE-PPO Model.

3.1 Data Collection

This study is grounded in a robust and diverse financial dataset, meticulously curated to capture the multidimensional nature of contemporary markets. Core price data—including open, high, low, close, and volume (OHLCV)—is sourced through the open-source `yfinance` API, which offers reliable and timely access to historical market data. To augment this raw data, a comprehensive set of technical indicators is computed based on OHLCV values. These include the Simple Moving Average (SMA), Exponential Moving Average (EMA), Raw Stochastic Value (RSV), and the Relative Strength Index (RSI), among others, enabling the model to capture various aspects of price momentum and trend dynamics. Beyond technical features, the dataset incorporates a wide range of macroeconomic indicators, capturing broader economic conditions such as employment levels, inflation rates, GDP growth, consumer sentiment, and labor productivity. These variables are retrieved via the `fredapi` interface from the Federal Reserve Economic Data repository. Given the lower frequency of many economic indicators (e.g., monthly or quarterly releases), time series are upsampled to a daily frequency using forward-fill imputation to ensure temporal consistency and continuity across all features. To ensure a globally informed modeling process, the dataset also integrates major international equity indices—including the SSE Composite Index, Nikkei 225, FTSE 100, and S&P 500—alongside sectoral fund indices and treasury yield curves (notably, the 5-year and 10-year U.S. Treasury yields). These series are treated as individual nodes in the financial graph, allowing the model to learn from diverse market signals and capture relationships between domestic and international financial instruments.

3.2 Data Pre-processing

Effective data pre-processing is critical to ensure the integrity and performance of the downstream learning models. Raw time series often contain missing values, inconsistencies, and outliers that must be addressed prior to training. Missing values in technical and macroeconomic indicators are handled using forward-fill imputation, preserving temporal dependencies without introducing artificial bias. Outliers are detected using statistical thresholds and treated appropriately to mitigate their potential impact on model stability and inference. The normalization is applied to all features, typically through z-score standardization using Equation (1):

$$z_i = \frac{x_i - \mu_x}{\sigma_x} \quad (1)$$

where z_i denotes the normalized value of the i -th feature, x_i is the original value of the i -th feature, μ_x represents the mean of the feature values, and σ_x is the standard deviation of those values. This transformation ensures that each feature is rescaled to have zero mean and unit variance, allowing variables with different units and scales to be meaningfully compared and processed by the model. This ensures that variables with different units and scales can be meaningfully compared and processed by the model. To further enhance the dataset, feature selection is performed using SHAP values. This approach identifies the most informative predictors, reducing the dimensionality of the data and improving both model efficiency and interpretability. The result is a clean, well-structured dataset that is ready for the construction of the financial graph.

3.3 Methods Used

3.3.1 Financial Graph Construction

To move beyond treating assets in isolation, a financial graph is constructed to explicitly model the relationships between different entities. In this graph, each node represents an asset, index, or economic indicator, while edges capture the statistical relationships—primarily correlations—between them. The correlation matrix is calculated across all pairs of nodes (see Figure 2) and edges are established between those with correlation coefficients exceeding a predefined threshold, as suggested by [Krauss Christopher \(2017\)](#). The resulting structure is encoded as both an adjacency matrix and a corresponding edge list, which together define the network structure of the financial market. By representing the market as a graph (see Figure 3), the model is able to leverage both local connections—such as sectoral or

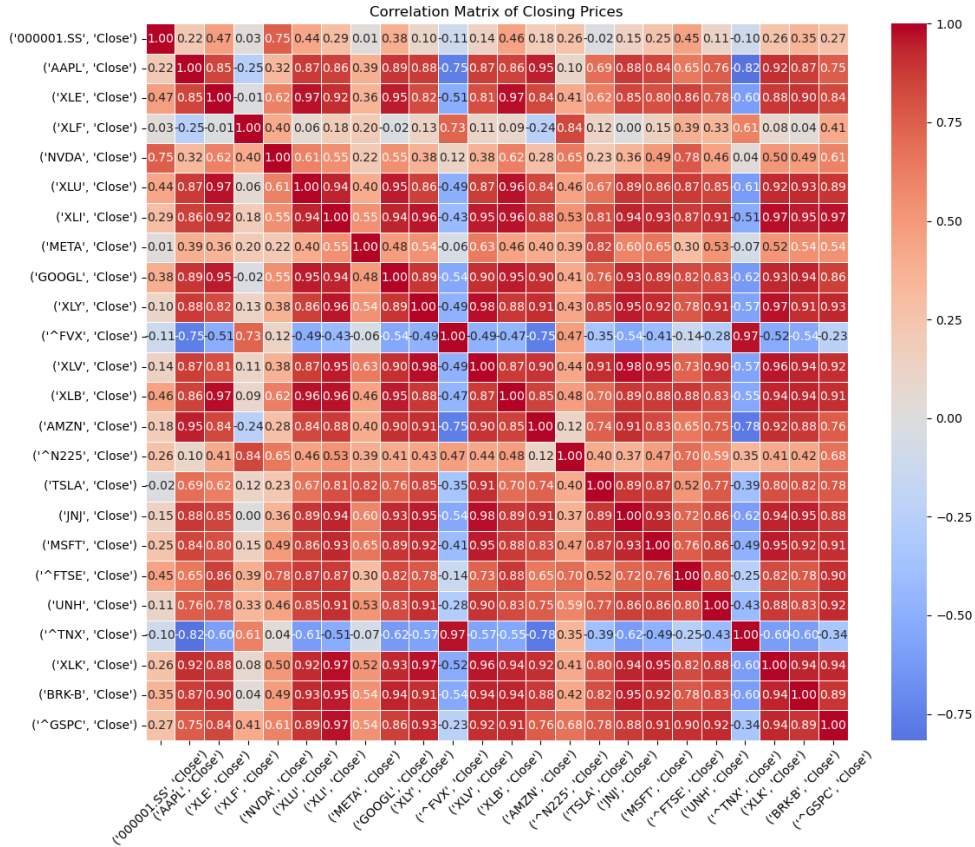


Figure 2: Correlation Matrix of Closing Prices to Construct the Financial Graph.

industry-specific relationships—and global patterns that reflect broader market movements. This graph-based representation provides a richer and more realistic foundation for subsequent analysis, enabling advanced models like GraphSAGE to aggregate information from each node's neighborhood and capture the complex, non-Euclidean dependencies that drive asset behavior. Such a structure not only enhances the model's ability to detect hidden patterns and systemic risk but also supports more robust and adaptive portfolio optimization strategies.

3.3.2 Feature Selection and Integration

In financial portfolio optimization, feature engineering plays a much deeper role than simply computing basic indicators or applying standard transformations. It involves carefully merging a variety of data sources to build a comprehensive, structured view of the market. In our framework, we bring together numerical features (such as historical asset prices and technical momentum indicators), macroeconomic variables, and index-based signals to create a high-dimensional feature matrix. These node-level features are then linked using graph structures—specifically, an adjacency matrix and edge list—so that each asset becomes part of a connected network. This graph-based representation enables the model to understand not only how individual assets behave but also how they relate to one another across time and economic conditions. By capturing both the intrinsic properties of assets and their relationships in the broader market context, we enable the graph neural network to learn more nuanced real-world patterns of financial interaction (Kamesh Korangi, 2024). To assess the relevance of each feature and enhance interpretability, we apply SHAP values, which help quantify the contribution of each variable to the model output. Specifically, SHAP values are computed by training a Gradient Boosting classifier on the feature matrix, where the target variable is defined as a binary outcome: 1 if the current closing price is greater than or equal to the average of the past five days, and 0 otherwise. This formulation allows the model to learn directional movements in price behavior. As shown in Figure 4, features like moving average convergence divergence, volatility, and interest rate data have higher SHAP values, indicating

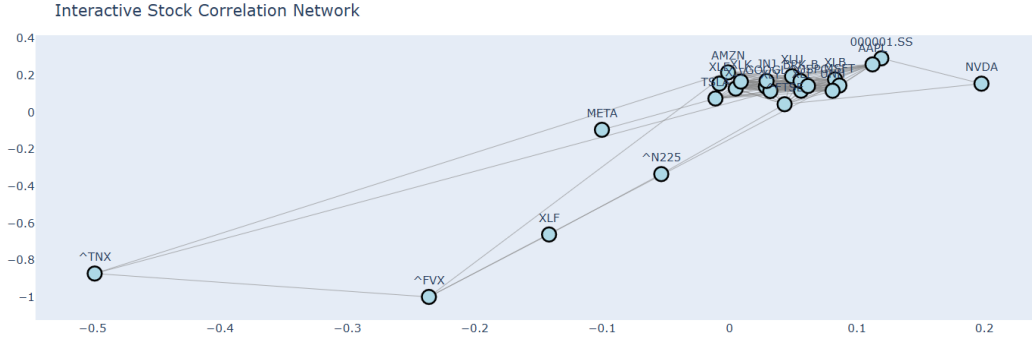


Figure 3: Financial Graph constructed using correlation values

their significant influence in the learning process [Scott M. Lundberg \(2017\)](#); [Qiguo Sun a \(2024\)](#).

In the domain of financial portfolio optimization, feature engineering transcends simple transformations or conventional technical indicators. It involves constructing an integrated, information-dense representation of the market environment. In this study, we unify multiple data sources into a graph-structured format by combining the feature matrix, adjacency matrix, and edge list. Each asset in the graph is associated with a comprehensive set of attributes, including historical price data, momentum-based indicators, macroeconomic variables, and co-movements with benchmark indices. This formulation enables the model to learn not only from asset-specific features but also from inter-asset relationships encoded in the market topology. To assess the relative importance of these features and enhance model interpretability, we employ SHapley Additive exPlanations values—a method rooted in cooperative game theory. The SHAP value ϕ_i for a given feature i is mathematically defined in Equation (2) :

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (2)$$

Here, F denotes the full set of input features, S is any subset excluding feature i , and $f(S)$ represents the model output when trained on S . This formulation computes the marginal contribution of each feature by averaging its impact across all permutations of feature subsets. Utilizing SHAP values serves a dual purpose. First, it offers model transparency by elucidating how specific features influence prediction outcomes—an essential requirement in financial applications subject to regulatory scrutiny. Second, it facilitates effective feature selection by identifying and excluding variables with low marginal utility. This improves the efficiency of the graph neural network by focusing learning capacity on high-impact signals, thereby enhancing both convergence and generalizability. By integrating this feature-level relevance with structural insights from the graph topology, the proposed framework is able to uncover latent, non-linear relationships in financial data that would otherwise remain inaccessible in non-relational learning paradigms [Scott M. Lundberg \(2017\)](#); [Cheng et al. \(2022\)](#).

3.4 Feature Extraction using GraphSAGE

Once the financial graph is constructed, we employ GraphSAGE, a state-of-the-art graph neural network architecture, to extract informative embeddings for each asset node ([Hamilton et al., 2017](#)). Unlike traditional feedforward networks, which operate on fixed-size input vectors, GraphSAGE is designed to learn from graph-structured data by aggregating information from a node's local neighborhood. This enables the model to encode both the intrinsic features of each asset and the topological context defined by its connected peers. At each layer k , the representation of a node v , denoted as $h_v^{(k)}$, is computed by aggregating the embeddings of its neighbors and combining them with its own previous-layer embedding defined in Equation (3) :

$$h_v^{(k)} = \sigma \left(W^{(k)} \cdot \text{CONCAT} \left(h_v^{(k-1)}, \text{AGGREGATE}^{(k)} \left(\{h_u^{(k-1)} : u \in \mathcal{N}(v)\} \right) \right) \right) \quad (3)$$

Here, $W^{(k)}$ is a trainable weight matrix for the k -th layer, σ is a non-linear activation function such as ReLU, and $\text{AGGREGATE}^{(k)}$ is a function like mean, max pooling, or LSTM-based sequence modeling.

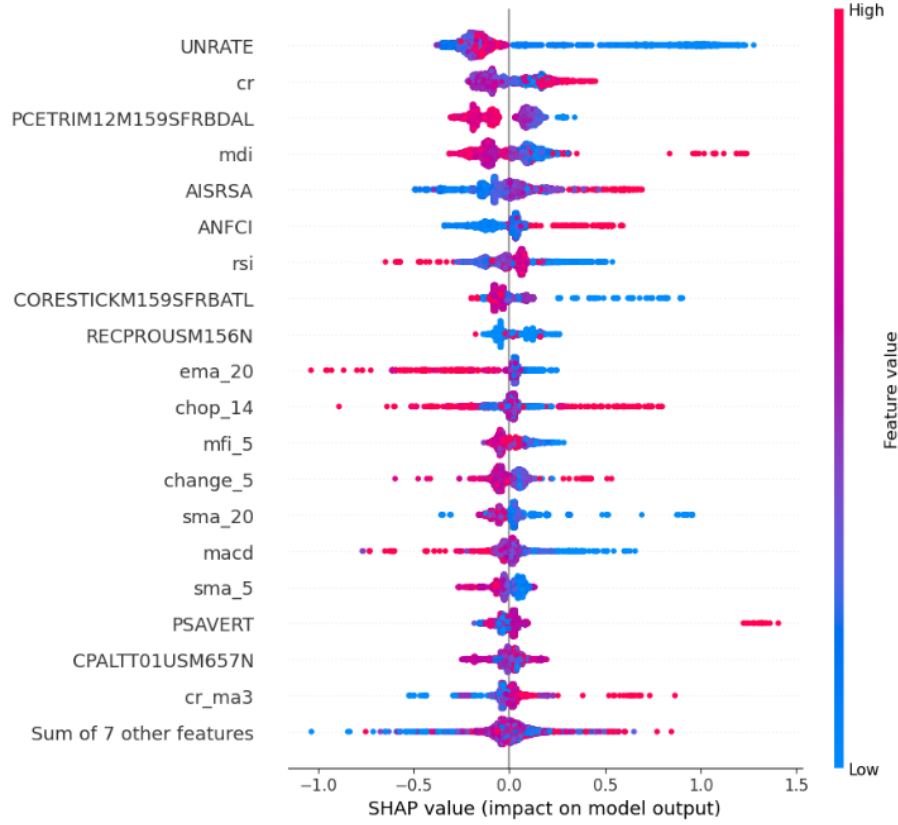


Figure 4: SHAP values of different features

The input to the first layer $h_v^{(0)}$ consists of asset-level features such as historical prices, technical indicators, and macroeconomic signals. Through multiple iterations of this aggregation, GraphSAGE produces compact, low-dimensional node embeddings that encode both local features and global structural dependencies. These embeddings are crucial for modeling market interactions, allowing the reinforcement learning agent to understand the contextual role of each asset within the broader network (Sun et al., 2024).

3.5 Embedding Generation

The embeddings produced by GraphSAGE serve as context-aware input representations for downstream reinforcement learning tasks. These embeddings integrate both static and dynamic properties of financial assets, capturing temporal indicators (e.g., moving averages, volatility) alongside relational information (e.g., sectoral proximity, co-movement with indices). Unlike raw feature vectors, these dense embeddings provide a more expressive and generalized view of the financial system. This representation allows the reinforcement learning agent to perceive market dynamics not just from the perspective of individual asset performance, but also from its embedded role within the financial ecosystem. The dimensionality reduction implicit in the GraphSAGE process further aids in regularization, improving the agent's ability to generalize across different market regimes while mitigating overfitting.

3.6 Training the PPO Model in Financial Environment

The core decision-making engine of the proposed framework is a Proximal Policy Optimization (PPO) agent trained within a custom-designed financial simulation environment. This environment mimics real-world trading dynamics, allowing the agent to interact with the market by making sequential

allocation decisions. At each time step t , the PPO agent receives as input a compact and informative state representation composed of node embeddings $\mathbf{s}_t$ generated by the GraphSAGE network. These embeddings encode both the individual characteristics of assets and their structural interdependencies within the financial market graph. Based on the input state $\mathbf{s}_t$, the agent produces an action vector $\mathbf{a}_t \in \mathbb{R}^N$, where each element in the vector corresponds to the allocation weight assigned to a specific asset among the N total assets in the portfolio.

To train the PPO agent effectively, we optimize a clipped surrogate objective that enhances training stability by preventing drastic updates to the policy. The clipped loss function is expressed as shown in Equation (4) :

$$\mathcal{L}_{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (4)$$

In this equation, θ denotes the current parameters of the policy network π_θ , and $r_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t|\mathbf{s}_t)}$ represents the ratio of the probabilities of the selected action $\mathbf{a}_t$ under the new and old policies, given the same state $\mathbf{s}_t$. The term $\hat{A}_t$ denotes the advantage estimate at time t , quantifying how much better the chosen action performed relative to the baseline value function. The advantage estimate $\hat{A}_t$ is computed using Generalized Advantage Estimation (GAE), which balances bias and variance by incorporating both short-term rewards and long-term value predictions. This is formalized as shown in Equation (5) :

$$\hat{A}_t = \sum_{l=0}^{T-t} (\gamma\lambda)^l \delta_{t+l}, \quad \text{where} \quad \delta_t = r_t + \gamma V(\mathbf{s}_{t+1}) - V(\mathbf{s}_t) \quad (5)$$

Here, $\gamma \in [0, 1]$ is the discount factor that controls the importance of future rewards, $\lambda \in [0, 1]$ is the smoothing parameter for GAE, and δ_t is the temporal-difference error at time t . The function $V(\mathbf{s}_t)$ denotes the value of the current state as estimated by the critic network, and r_t is the immediate reward received after executing action $\mathbf{a}_t$. The complete loss function optimized during training includes not only the clipped surrogate objective but also a value function loss and an entropy regularization term to promote exploration. The total PPO objective is shown in Equation (6):

$$\mathcal{L}_{\text{TOTAL}} = \mathcal{L}_{\text{CLIP}} + c_1 \mathcal{L}_{\text{VALUE}} - c_2 \mathcal{L}_{\text{ENTROPY}} \quad (6)$$

In this expression, $\mathcal{L}_{\text{VALUE}}$ measures the discrepancy between the predicted and actual returns, typically via mean squared error, and $\mathcal{L}_{\text{ENTROPY}}$ quantifies the entropy of the policy distribution to encourage randomness in action selection. Coefficients c_1 and c_2 are hyperparameters that control the trade-off between accurate value estimation and exploratory behavior. A critical innovation in our PPO-based approach is the integration of a dynamic risk control mechanism designed to improve model robustness during periods of sustained underperformance. If the portfolio experiences a significant drawdown—defined as a percentage decline in portfolio value over a specified lookback window—the model automatically shifts to an equal-weight allocation strategy to mitigate further losses. This fallback mechanism is governed by Equation (7):

$$\text{If } \frac{V_t - V_{t-n}}{V_{t-n}} < -\alpha, \quad \text{then } \mathbf{a}_t = \left[\frac{1}{N}, \frac{1}{N}, \dots, \frac{1}{N} \right] \quad (7)$$

In this formulation, V_t is the current portfolio value, V_{t-n} is the portfolio value n timesteps ago, $\alpha \in (0, 1)$ is a predefined threshold representing acceptable drawdown (e.g., 0.05 for 5%), and $\mathbf{a}_t$ is the resulting portfolio allocation vector. The equal-weight allocation ensures that each asset receives $\frac{1}{N}$ of the capital, serving as a conservative, risk-neutral fallback. This reversion strategy emulates practical risk management techniques employed by institutional investors and serves as a fail-safe mechanism when the agent's policy deteriorates in highly volatile or unpredictable environments. By incorporating this adaptive control into the training loop, the framework not only improves long-term performance but also enhances its robustness and interpretability—two essential properties in real-world financial decision-making.

4 Case Study

4.1 Background

In modern financial markets, investors continuously grapple with the challenge of allocating capital across a diverse universe of assets. These assets are interconnected through a complex mesh of economic, sectoral, and global dependencies, which traditional machine learning and deep learning models often fail to represent effectively. While recent reinforcement learning approaches offer adaptability to evolving market conditions, many of them operate in a univariate or asset-isolated framework and lack integrated mechanisms for risk-aware decision-making.

This study introduces a hybrid framework that addresses these limitations by fusing graph neural networks with deep reinforcement learning. A financial graph is constructed in which each node represents an individual asset or market index, and edges encode relationships such as significant price correlations or shared macroeconomic influences. Using the GraphSAGE architecture, we generate low-dimensional embeddings that encapsulate not only the intrinsic characteristics of each asset but also contextual information from its connected neighbors within the graph structure. These embeddings are subsequently utilized by a Proximal Policy Optimization agent, which learns a dynamic asset allocation policy based on evolving market signals. This graph-based RL framework offers enhanced structural awareness and adaptability, aiming to produce more resilient and stable portfolio performance than conventional approaches.

4.2 Data Collection and Description

To faithfully model the intricacies of real-world financial behavior, a comprehensive and heterogeneous dataset was curated from multiple publicly accessible sources. The primary layer of data comprises asset-level OHLCV information, retrieved via the `yfinance` API. This ensures timely and accurate data coverage across a broad spectrum of equities. To enrich this foundational dataset, technical indicators such as the Simple Moving Average, Exponential Moving Average, Raw Stochastic Value, and Relative Strength Index were computed using the `stockstats` library. These indicators help the model capture short-term and long-term trends, overbought/oversold conditions, and momentum shifts essential for trading and allocation strategies. Complementing these technical signals, macroeconomic variables were integrated to provide a holistic view of market forces. Key indicators—including employment data, inflation rates, GDP growth, consumer sentiment, and productivity metrics—were extracted from the Federal Reserve Economic Data (FRED) via the `fredapi`. Given the lower reporting frequency of such variables, they were upsampled to a daily resolution and imputed using forward-fill techniques to align with the frequency of financial market data.

To introduce a global and sectoral dimension to the dataset, additional time series were included for international stock indices such as the SSE Composite Index, Nikkei 225, FTSE 100, and S&P 500, as well as sector-specific exchange-traded funds and U.S. Treasury yields for 5-year and 10-year bonds. These were treated as nodes within the constructed graph, enabling the learning model to incorporate inter-market relationships and macro-level dynamics. This multi-source, multi-scale data architecture equips the model to interpret both asset-specific behaviors and higher-order financial interactions, thereby supporting more informed and context-aware portfolio optimization.

5 Results and Discussions

The evaluation of the proposed GraphSAGE-PPO framework yields valuable insights into its efficacy for real-world portfolio optimization. When benchmarked against both the Equal Weight strategy and a standard PPO baseline, the model consistently outperformed across key return and risk-adjusted metrics. As shown in Figure 7, the cumulative portfolio value generated by the GraphSAGE-PPO agent followed a consistently upward trajectory, outperforming both baselines across the entire test horizon. Notably, the performance gap widened during periods of increased market volatility, underscoring the model's ability to capture short-term opportunities while sustaining long-term growth. This highlights the

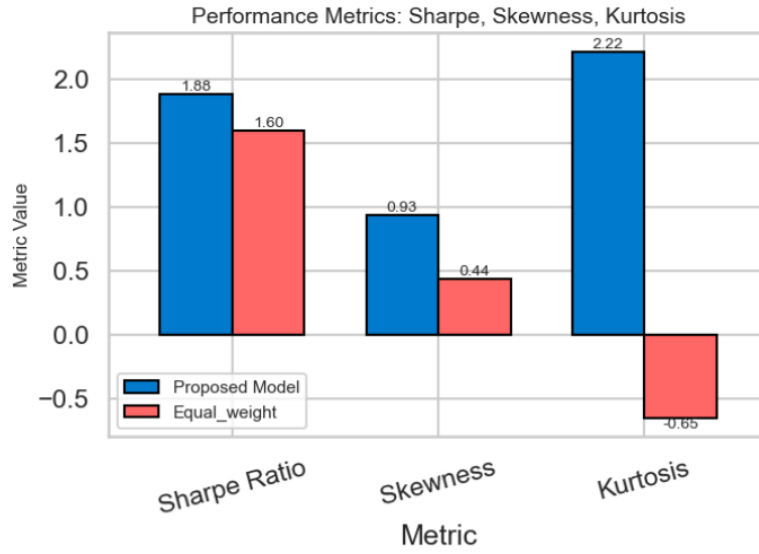


Figure 5: Performance Metrics: Sharpe, Skewness, Kurtosis

advantage of incorporating relational financial structures into the learning process to support dynamic, data-driven decision-making. Figures 5 and 6 provide a deeper view into the model’s statistical and risk characteristics. Figure 5 focuses on Sharpe ratio, skewness, and kurtosis—metrics that reflect return quality and distribution symmetry. The proposed model achieved a significantly higher Sharpe ratio, indicating stronger risk-adjusted returns, along with skewness and kurtosis values that suggest a more balanced and stable return profile. Figure 6 compares volatility, win rate, and tail risk at the 5th percentile. While both models demonstrated similar win rates, the GraphSAGE-PPO agent recorded lower tail risk and reduced return volatility. These results indicate a smoother performance curve and improved downside protection, both of which are highly desirable in institutional portfolio management.

The model also demonstrated strong adaptability to changing market dynamics. Its ability to adjust asset allocations in response to shifts in macroeconomic indicators and price movements emerged organically from the training process. This adaptive behavior reflects the agent’s learned capacity to interpret both feature-level inputs and structural signals within the asset graph. Such context-aware rebalancing enables the agent to navigate varying market conditions more effectively than rigid, rule-based systems. However, one noted limitation is the absence of an explicit risk-sensitive component in the reward function. While the model achieves favorable risk-adjusted returns overall, the lack of formal penalization for high volatility or drawdowns occasionally results in abrupt performance dips during sharp market corrections. This represents a promising area for future enhancement through the integration of risk-sensitive reward shaping or constraint-aware optimization strategies.

5.1 Factors Interpretation

The key factors influencing portfolio performance in our study stem from the model’s ability to learn both the unique characteristics of each asset and the broader network of relationships within the market. As highlighted in Table 2, the use of GraphSAGE for graph-based feature extraction allowed the agent to capture complex dependencies between assets—such as sector ties or shared exposure to economic indicators—that traditional models often overlook. This relational learning enabled the model to recognize patterns of co-movement or risk clustering, which proved especially valuable during periods of market volatility.

The integration of these graph-derived embeddings with the adaptive learning capabilities of the PPO agent resulted in a more responsive and resilient portfolio management strategy. The agent was able to dynamically adjust asset allocations in response to shifting market conditions, as evidenced by improved evaluation matrices compared to baseline approaches. Notably, the model’s ability to reallocate towards

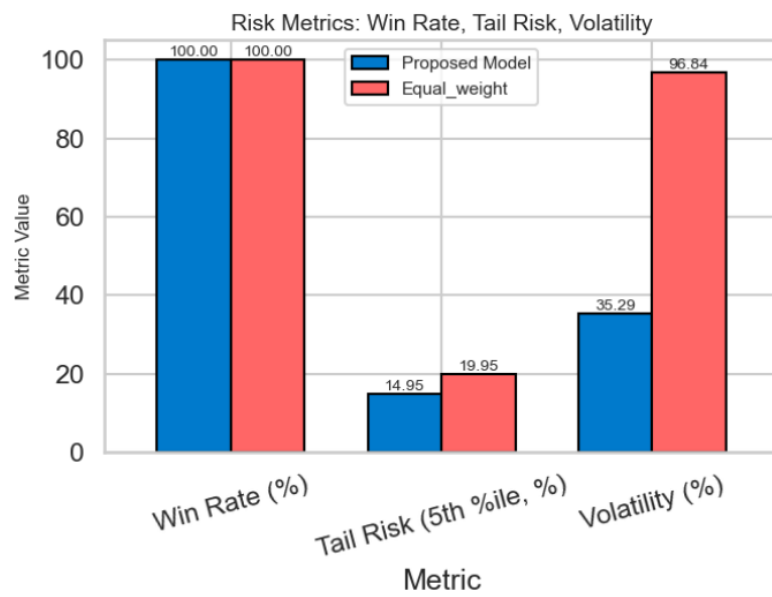


Figure 6: Risk Metrics: Win Rate, Tail Risk, Volatility

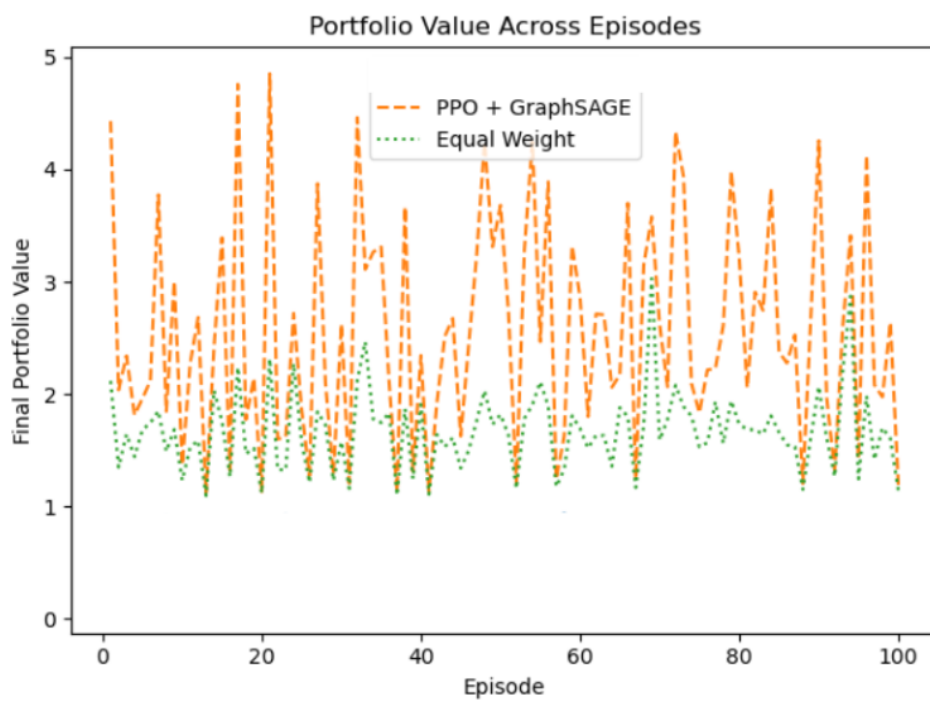


Figure 7: Portfolio Value for after 100 episodes.

defensive sectors during turbulent periods demonstrated a practical advantage in real-world scenarios. Overall, these findings underscore the importance of combining both asset-specific features and relational context in portfolio optimization. By leveraging graph-based learning, our approach not only enhanced profitability but also improved risk management and adaptability—key factors that drive long-term investment success.

Table 2: Summary of Key Model Components and Their Portfolio Impact

Factor	Interpretation	Key Insights from Analysis
Graph-based Feature Extraction	Use of GraphSAGE to encode asset relationships	Enabled the model to capture complex market dependencies
Adaptive Portfolio Allocation	PPO agent trained on GraphSAGE embeddings	Led to dynamic, data-driven adjustments in asset allocation
Risk-adjusted Performance	Higher Sharpe/Sortino ratios, lower drawdown	Demonstrated improved risk management and portfolio resilience
Market Responsiveness	Allocation shifts during volatility	Model reallocated to defensive sectors in response to shocks
Robustness and Generalizability	Consistent outperformance over benchmarks	Outperformed Equal Weight and S&P 500 across multiple datasets

5.2 Dominance Analysis and Dominant Factors Influencing Customer Satisfaction in Portfolio Management

In the context of portfolio management, several dominant factors have been identified as key drivers of customer satisfaction. Foremost among these is the reliability and consistency of returns; clients place significant value on a portfolio management model's ability to deliver stable performance over time, often benchmarking against indices like the S&P 500. Such reliability is fundamental for building investor trust and fostering long-term satisfaction (Dangalla, 2024). Equally important is the model's responsiveness and adaptability—its capacity to swiftly adjust asset allocations in response to market volatility or macroeconomic shifts reassures clients that their investments are being actively and intelligently managed. Transparency and explainability further enhance client confidence. When portfolio decisions are accompanied by clear, interpretable rationales—such as those provided by SHAP value analyses or other explainable AI tools—clients are more likely to understand and trust the investment process (Scott M. Lundberg, 2017; Sun et al., 2024). Personalization and customization also play a critical role; tailoring strategies to individual goals, risk appetites, and financial circumstances not only meets client needs but also strengthens the client-advisor relationship (Stefan Colza Lee, 2018). Effective risk management is another cornerstone of satisfaction, with clients expecting proactive measures to minimize drawdowns and protect capital during market downturns. Demonstrating robust risk controls is essential for client peace of mind, especially in uncertain or turbulent markets (Stefan Colza Lee, 2018). Finally, ease of use and accessibility—through intuitive interfaces, clear reporting, and seamless access to portfolio information—contributes significantly to a positive client experience. Clients increasingly value platforms that make it straightforward to monitor, understand, and interact with their investments, reflecting a broader trend toward user-centric financial technology solutions (Scott M. Lundberg, 2017).

5.3 Theoretical Contributions

This work significantly advances portfolio management research by bridging graph theory with deep reinforcement learning, addressing critical limitations in traditional Modern Portfolio Theory and its derivatives. While MPT relies on static correlations and Euclidean data assumptions, our framework introduces a dynamic, relationally-aware approach through GraphSAGE embeddings. By explicitly modeling non-Euclidean interdependencies among assets—such as sectoral ties, index co-movements,

and macroeconomic influences—the study extends the Capital Asset Pricing Model and multi-factor frameworks, which historically treat assets in isolation. The integration of GraphSAGE with Proximal Policy Optimization advances DRL applications in finance, addressing the "curse of dimensionality" in high-dimensional financial datasets. Unlike conventional DRL models that focus on temporal patterns alone, our architecture captures both structural market relationships and temporal dynamics, aligning with emerging theories of network-driven market behavior. This hybrid approach also contributes to explainable AI in finance by using SHAP values to quantify feature importance, thereby connecting black-box DRL decisions to interpretable economic indicators—a critical step toward meeting regulatory and institutional transparency demands.

5.4 Practical implications

The proposed GraphSAGE-PPO framework provides a powerful toolkit for modern asset managers, institutional investors, and fintech platforms. By embedding asset-level features into a graph structure and learning from both node attributes and inter-asset relationships, the model captures complex dependencies that traditional methods like correlation matrices often miss. This enhanced relational understanding supports more effective diversification, helping reduce systemic risk and build portfolios that are more robust to market volatility and macroeconomic shocks. The framework's ability to integrate real-time market signals and dynamically reallocate capital gives it an edge over static rebalancing strategies, especially during periods of uncertainty or regime shifts. Scalability and adaptability are also central advantages of this approach. The use of low-dimensional GraphSAGE embeddings makes the system efficient even with large asset universes, making it suitable for deployment in institutional funds, robo-advisory services, and high-frequency trading environments. By enabling context-aware, data-driven allocation decisions, the framework can help reduce turnover, minimize slippage, and lower transaction costs. Its architecture also allows for the integration of custom constraints and investor preferences—such as ESG criteria, liquidity screens, or sentiment-driven signals—thereby increasing its applicability to a wide range of investment mandates.

Despite these promising characteristics, deploying the system in live financial environments requires careful calibration. Real-world implementation should ideally begin with sandbox testing or limited-scale pilots, followed by the integration of risk-control mechanisms such as volatility thresholds, transaction cost models, and scenario-based stress testing. These steps ensure that the model remains compliant, interpretable, and responsive under live trading conditions. Altogether, the GraphSAGE-PPO framework exemplifies how advanced AI can be grounded in practical portfolio workflows, offering both cutting-edge performance and operational readiness for next-generation asset management.

5.5 Limitations and Future Work

While the proposed Graph Reinforcement Learning framework shows considerable promise in advancing portfolio management, several limitations remain that offer valuable directions for future research. One of the most significant limitations is the static nature of the financial graph used in the current model. Throughout the training process, the graph structure remains fixed, failing to account for temporal shifts and evolving relationships among assets—such as changes due to sector realignments, macroeconomic cycles, or market regime transitions. This static representation limits the model's adaptability in real-world, dynamic financial environments. Recent developments in dynamic graph neural networks, including Temporal Graph Attention Networks and update-propagation frameworks, suggest that incorporating time-varying structure into the learning process can significantly enhance a model's ability to capture market evolution and volatility patterns (Pulikandala et al., 2024). Future extensions of this work should explore dynamic graph construction techniques and temporal GNNs to improve the responsiveness and realism of the modeling framework.

Another area that warrants attention is the model's current approach to hyperparameter tuning. The use of manual, trial-and-error techniques is not only computationally inefficient but also suboptimal in discovering globally effective configurations. Automated and principled optimization methods, such as Bayesian optimization, population-based training, or adaptive grid search, have been shown to produce

more reliable and reproducible outcomes in reinforcement learning systems (Carmona Pedrajas, 2023). Incorporating such techniques could substantially enhance the stability and performance of the GRL framework, especially when scaling to larger portfolios or adapting to new market regimes. Additionally, the model relies exclusively on Proximal Policy Optimization for decision-making. While PPO is known for its robustness and simplicity, recent studies indicate that alternative DRL algorithms such as Soft Actor-Critic, Twin Delayed Deep Deterministic Policy Gradient, and Asynchronous Advantage Actor-Critic may offer superior exploration capabilities and stability, particularly in noisy or high-dimensional financial environments (Haarnoja et al., 2018; Nigel Cuschieri, 2025). Investigating these alternatives in combination with graph-based representations could yield deeper insights and further strengthen the overall framework. Looking forward, several enhancements can be pursued to advance this research. These include implementing automated hyperparameter optimization strategies to fine-tune learning rates, network architectures, and embedding dimensions; experimenting with different DRL backbones integrated with GraphSAGE or alternative GNNs; and adopting time-evolving graph models with temporal edge predictors to better capture market fluctuations. Moreover, testing the framework on a broader set of asset classes—such as commodities, cryptocurrencies, and fixed-income securities—as well as in emerging markets, would be essential for evaluating its generalizability. Lastly, efforts to improve computational efficiency by leveraging lightweight graph architectures, such as Graph Attention Networks or simplified neighborhood aggregation techniques, could make the model suitable for real-time trading scenarios and applications at institutional scale (Shibo Feng a, 2022). Taken together, these future directions highlight a clear pathway for transforming GRL-based systems into powerful, scalable, and adaptive solutions for next-generation financial portfolio optimization.

6 Conclusions

This study set out to address enduring challenges in portfolio optimization, particularly the absence of structural awareness in conventional deep reinforcement learning models and the limitations of traditional machine learning approaches in capturing the intricate, dynamic interdependencies among financial assets. To overcome these limitations, we proposed a novel hybrid framework that combines GraphSAGE—a graph neural network architecture designed to model relational dependencies among nodes—with Proximal Policy Optimization, a DRL algorithm well-suited for continuous decision-making tasks such as portfolio allocation. The agent was trained on a graph-structured representation of market data, enriched with diverse asset-level features including price movements, technical indicators, and macroeconomic variables. By embedding this high-dimensional information into a relational context, the framework enabled the agent to make informed, structure-aware portfolio decisions that accounted not only for individual asset dynamics but also for their positioning within the broader financial network. Empirical evaluations demonstrated that the proposed GraphSAGE-PPO model significantly outperformed a naive equal-weight strategy in terms of cumulative returns, thereby showcasing its potential for enhanced capital growth. However, this performance gain came with increased volatility and higher exposure to tail risk, suggesting a trade-off between return maximization and portfolio stability. Notably, while the equal-weight strategy exhibited superior risk-adjusted characteristics—reflected in higher Sharpe-like metrics and more favorable skewness and kurtosis—the proposed model showed greater adaptability to market structure and temporal changes. This adaptability is particularly valuable in capturing investment opportunities that static or feature-agnostic strategies may overlook. These findings highlight the importance of incorporating structural financial knowledge into learning agents, confirming that graph-aware reinforcement learning models offer a promising path toward building intelligent, resilient, and adaptive financial decision-making systems.

Future research can build upon this work in several impactful directions to further enhance the robustness, adaptability, and practical relevance of AI-driven portfolio optimization frameworks. First, exploring alternative deep reinforcement learning algorithms—such as Soft Actor-Critic, Twin Delayed Deep Deterministic Policy Gradient, or Distributional RL—could lead to more efficient policy learning by achieving better trade-offs between exploration and exploitation, while offering improved convergence under uncertainty and complex reward structures. These models may be particularly beneficial in capturing long-tail risks and nonlinear payoffs that are common in financial markets.

Second, integrating diverse Graph Neural Network architectures—including Graph Attention Networks, Temporal Graph Networks, and Heterogeneous GNNs—has the potential to better model evolving relationships between assets over time, especially in the presence of asynchronous information flows and multi-type dependencies. Such models can also capture temporal heterogeneity and regime shifts more effectively than static graph encoders. Additionally, hyperparameter optimization through techniques such as Bayesian Optimization, Grid Search, or Evolutionary Algorithms can significantly improve both the learning efficiency and generalizability of the model across different market regimes and asset classes. Another promising direction involves incorporating sentiment-aware decision-making by leveraging unstructured data from real-time news streams, analyst reports, and social media platforms (e.g., Twitter, Reddit), using natural language processing techniques to generate predictive sentiment signals. These inputs can complement technical indicators and price-based features, enabling the model to anticipate market-moving events and shifts in investor behavior. Moreover, the current reliance on a static correlation graph limits the model's adaptability; employing dynamically updated graphs—constructed from rolling-window correlations, mutual information metrics, or real-time sentiment co-movements—can improve responsiveness to abrupt structural changes and systemic shocks. Furthermore, the framework could be extended to support multi-asset and cross-market portfolios by including asset classes such as fixed income securities, foreign exchange instruments, commodities, and even cryptocurrencies. This would not only diversify the portfolio further but also make the framework more suitable for institutional asset managers with broader mandates. Finally, future enhancements could include multi-objective reward functions and constraint-based policy optimization, allowing the system to align more closely with real-world investment guidelines, such as ESG constraints, liquidity limits, and regulatory capital requirements. By addressing these extensions, the proposed framework could evolve into a more comprehensive and resilient financial decision-support system capable of navigating a wide range of market environments.

Conflict of Interest

The author declares that there is no conflict of interest in this work.

Data Availability

Data may be available on request.

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